

Tables II–III Value Provenance Audit

Paper: `paper/6a8e554fac122323bf4b0f87/root.tex`

Audited sections: Experiment 2 (Table II) and Experiment 3 (Table III)

Audit date: September 13, 2026

Result. All displayed numerical values in Tables II and III match the current analysis outputs after the rounding used in the paper. Table II objective inferential statistics were independently recomputed from the trial-level CSV and also match. The principal provenance caveat is that no original Study 2 script exported those objective tests; the accompanying `audit_experiment_tables.py` now reproduces them. One Experiment 3 procedure/count distinction requires clarification, described below.

How internally calculated values are produced

- **Mean and SD:** trials/steps are first summarized within participant and condition; the table reports the mean and sample SD across participants.
- **Friedman test:** three paired condition vectors are compared; Table II objective tests use $N=10$ and questionnaire tests use $N=11$.
- **Kendall's W:** `Friedman chi-square / [N × (3-1)]`.
- **Pairwise prose values:** paired, two-sided Wilcoxon signed-rank tests followed by Holm correction across the three condition pairs for each metric.

Table II — Experiment 2 objective metrics

Primary data: `study_logs/study2/study2_efficiency_conditions.csv`

Extractor: `study2_replay_to_csv.py`

Independent statistical reproduction: `paper_related/audit_experiment_tables.py`

The extractor truncates every trial at first target reach, retains trials 0–7, applies edited grasp/orientation-noise annotations, and outputs one row per participant × assigned condition × trial. For each table cell, the relevant CSV column is averaged across trials within each participant; the ten participant means are then averaged and their sample SD calculated.

Paper row	CSV column	Freedrive	AR Handle	Hybrid	Recomputed p / W
Right-hand path (m)	<code>hand_movement_m</code>	2.0360 ± 0.6525	1.5031 ± 0.2007	1.2360 ± 0.1472	0.000225 / 0.840
Interaction time (s)	<code>interaction_time_s</code>	9.2611 ± 3.4657	9.6741 ± 4.2871	5.8398 ± 2.1446	0.000553 / 0.750
Completion time (s)	<code>completion_time_s</code>	10.4772 ± 4.4081	20.9606 ± 5.7574	15.4462 ± 2.9596	0.000112 / 0.910
Interactions (#)	<code>num_interactions</code>	2.7625 ± 1.3143	1.8411 ± 0.2931	1.7000 ± 0.3689	0.041411 / 0.318
Translation gain/hand (m/m)	<code>translation_per_hand_m</code>	0.4954 ± 0.1123	0.5928 ± 0.0698	0.6714 ± 0.0735	0.000676 / 0.730
Rotation gain/hand (deg/m)	<code>rotation_per_hand_m</code>	47.9635 ± 12.0784	50.7794 ± 6.4564	61.2040 ± 6.1186	0.002243 / 0.610

All six rows round to the values printed in Table II.

Definitions originating inside the Study 2 extractor

Metric	Internal origin
Right-hand path	Cumulative Euclidean displacement of valid right-hand samples from the first interaction to first target reach, after discontinuity filtering. Orientation-noise hand movement is removed for the AR-affected channel.

Interaction time	Union duration of active interaction intervals: annotated AR grasp–release intervals and detected Freedrive robot-motion intervals. Overlap in Hybrid is counted once.
Completion time	Elapsed time from the first interaction (first robot movement for Freedrive or first annotated AR grasp) to first maintained target reach.
Number of interactions	Count of AR grasp episodes and/or Freedrive motion episodes before first reach.
Translation gain/hand	Translation-error reduction divided by right-hand path length.
Rotation gain/hand	Rotation-error reduction divided by right-hand path length. Annotated user-orientation mistakes do not contribute a negative rotation penalty to the affected method.

Table II objective pairwise values supporting the text

Claimed comparison	Holm p from recomputation	Status
Completion: Freedrive vs AR; Freedrive vs Hybrid	.00586; .00586 → reported .006	Matches
Hand path: Hybrid vs Freedrive; Hybrid vs AR	.00586; .00586 → reported ≤.006	Matches
Interaction time: Hybrid vs Freedrive; Hybrid vs AR	.00586; .00586 → reported .006	Matches
Translation gain/hand: Hybrid vs Freedrive; Hybrid vs AR	.00586; .03906 → reported .006 and .039	Matches
Rotation gain/hand: Hybrid vs Freedrive; Hybrid vs AR	.00781; .00586 → reported .008 and .006	Matches

Table II — Experiment 2 questionnaire metrics

Input: `paper_related/H2R_UserStudy_Analysis - study2_responses_side_by_side.csv`

Analysis: `paper_related/format_qualtrics_condition_comparison.py`

Descriptive output: `paper_related/study2_edited_condition_summary.csv`

Inferential output: `paper_related/study2_edited_pairwise_tests.csv`

Readable report: `paper_related/study2_edited_analysis_report.md`

Paper row	Output scale name	Freedrive	AR Handle	Hybrid	p / W
Workload	Task-Specific Workload	38.84 ± 18.30	35.64 ± 24.06	26.80 ± 16.36	.231 / .13
Adj. control	Adjustment Controllability	4.70 ± 0.99	5.27 ± 1.17	6.21 ± 0.70	<.001 / .69
Motion predict.	Robot Motion Predictability	6.21 ± 0.69	4.82 ± 1.26	5.67 ± 1.09	.005 / .48
Op. safety	Perceived Operational Safety	6.36 ± 1.14	5.64 ± 1.12	5.95 ± 0.91	.034 / .31
Position trust	Trust in Safe Positioning	5.82 ± 1.47	5.55 ± 1.21	5.82 ± 0.87	.687 / .03
Preference rank	Overall Preference Rank	2.55	2.27	1.18	.003 / .52

The workload composite averages Mental Demand, Physical Demand, Effort, and Frustration. Temporal Demand and Performance are excluded. Other constructs average the questionnaire items specified in the analysis script. Preference ranks are directly taken from the ranking response and then analyzed like the other paired outcomes.

The Table II questionnaire values and every pairwise statement in its results text match the current report.

Table III — Experiment 3 objective metrics

Raw input: `study_logs/study4/<participant>/<condition>.csv`

Extractor: `paper_related/study4_full_report.py`

Exact table source: `paper_related/study4_condition_summary.csv`

Conversation-level source: `paper_related/study4_conversations.csv`

Paper row	Exact output metric/formula	Gesture	Language	Task-Aware
Total time (s)	<code>total_time_s</code>	247.367 ± 111.561	239.389 ± 53.991	213.390 ± 50.144
Interactions (#)	<code>interactions_total</code> : hover rows for Gesture; VLM request/answer interaction rows for language conditions	274.182 ± 144.560	21.091 ± 3.177	21.000 ± 1.732
Wrong selections (%)	<code>100 × error_rate</code> : wrong selection events / all selection events, computed within participant before group mean/SD	19.916 ± 9.931	0.000 ± 0.000	0.000 ± 0.000
Head translation (m)	<code>head_translation_total_m</code>	10.492 ± 7.644	5.152 ± 3.422	4.786 ± 4.101
Head rotation (deg)	<code>head_rotation_total_deg</code>	2159.566 ± 1772.844	1689.411 ± 1216.630	1305.421 ± 926.154
Hand travel (m)	<code>total_hand_translation_total_m</code> = left + right hand	18.736 ± 12.055	9.560 ± 10.180	7.398 ± 10.214

These values round exactly to Table III.

Grounding-issue percentages

These are calculated from `study4_conversations.csv`, not from `study4_condition_summary.csv`:

- **Language 3.9%:** 9 issue turns / 232 total Language turns = 3.879%.
- **Task-Aware 5.2%:** 12 issue turns / 231 total Task-Aware turns = 5.195%.
- **Task-Aware D3-specific 24.7%:** 57 task-state query/cycle turns / 231 total Task-Aware turns = 24.675%. The 57 comprise `selected_step_physical_cycle` (33), `selected_step_needs_highlight` (18), `selected_step_object_already_acquired` (4), and `step_cycle_empty` (2).

`study4_conversation_review.py` exports non-clean turns for inspection, but the 3.9% and 5.2% paper values follow the decision-category grouping above. The 0% wrong-selection values are structurally different: selection is logged only after

language grounding resolves.

Procedure/count clarification needed. The paper says participants completed “the same seven matched partial-assembly steps.” The raw Study 4 configuration in `study4_full_report.py` contains eight step IDs: six bearing/gear-rod steps and two fastening instances. The script collapses the two fastening instances into one *reporting group* for per-step summaries, yielding seven groups, but the Table III per-session totals include the raw completed steps (generally eight; some sessions contain seven). Therefore, describe this as “eight task instances grouped into seven step types,” or recalculate the totals using exactly seven instances.

Table III — Experiment 3 questionnaire metrics

Input: `paper_related/H2R_UserStudy_Analysis - study4_responses_side_by_side.csv`

Analysis: `paper_related/format_qualtrics_condition_comparison.py`

Descriptive output: `paper_related/study4_edited_condition_summary.csv`

Inferential output: `paper_related/study4_edited_pairwise_tests.csv`

Readable report: `paper_related/study4_edited_analysis_report.md`

Paper row	Output scale name	Gesture	Language	Task-Aware	p / W
NASA-TLX	NASA-TLX	44.36 ± 26.70	27.52 ± 13.77	15.35 ± 11.35	.010 / .42
Assist. conf.	Assistance Confidence	5.09 ± 2.02	4.86 ± 1.36	6.00 ± 1.10	.214 / .14
Request ease	Request Clarity and Ease	4.14 ± 2.01	5.09 ± 1.28	6.18 ± 0.78	.013 / .39
Adapt. burden	Interaction Adaptation Burden	4.09 ± 2.12	3.18 ± 1.03	2.77 ± 1.94	.092 / .22
Preference rank	Overall Preference Rank	2.55	2.45	1.00	<.001 / .75

NASA-TLX averages all six dimensions after reverse-scoring Performance. Gesture excludes the inapplicable “system understood” item from Assistance Confidence and the inapplicable phrasing item from Adaptation Burden. Consequently, Gesture’s two composites are asymmetric and the paper appropriately cautions against strong cross-condition interpretation.

Experiment 3 text verification

Statement	Underlying number	Status
Gesture workload exceeds Language and Task-Aware	Holm p=.04883 and .03809	Reported .049 and .038
Request-ease comparisons with Task-Aware narrowly miss significance	Both Holm p=.05273	Reported .053
All participants rank Task-Aware first	11/11 rank 1; Friedman W=.752	Matches
Gesture has about 13× more interactions	274.182 / 21.091 = 13.00; / 21.000 = 13.06	Matches, but interaction units differ
Grounding issues are 3.9% and 5.2%	9/232 and 12/231	Matches category-based calculation
24.7% of TA turns invoke D3-specific behavior	57/231 = 24.675%	Matches

Reproduction commands

Study 2 extraction:

```
python3 study2_replay_to_csv.py study_logs/study2/*_replay.jsonl \
  --exclude mahya --csv study_logs/study2/study2_efficiency.csv
```

Study 2 objective verification:

```
/home/skim3674/anaconda3/envs/parakeet/bin/python -u \
  paper_related/audit_experiment_tables.py
```

Questionnaires:

```
/home/skim3674/anaconda3/envs/parakeet/bin/python \
  paper_related/format_qualtrics_condition_comparison.py
```

Study 4 totals and conversation extraction:

```
python3 paper_related/study4_full_report.py
```

Study 4 normalized behavioral tests:

```
/home/skim3674/anaconda3/envs/parakeet/bin/python \  
paper_related/study4_behavioral_analysis.py
```

Interpretation note: Table III intentionally leaves objective p/W blank. The separate behavioral-analysis script provides per-step normalized inferential comparisons, whereas the displayed table reports per-session totals. Do not copy those p-values into the total table without explicitly stating the change in analysis unit.